Development of an Online Tool and Acute Myeloid Leukemia Gene Expression Database for Biomarker Evaluation

Abstract:

Gene expression data analysis of an individual's cancer allows for a thorough understanding of the associations between their genetics and the clinical aspects of their cancer. The discovery of these gene associations, known as biomarkers, is incredibly important for furthering cancer treatment research. Acute Myeloid Leukemia (AML) is the most common form of aggressive leukemia in adults and is one of the deadliest forms of leukemia, with the average five-year survival rate across all subtypes only being around 30%. The high variability of subtypes is largely dependent on genes and their expression. *AML-BET* is a proposed web application, run using *R Shiny* and *ShinyProxy* using a *Docker* image for deployment, meant for the analysis of gene expression data in AML. By including gene expression data from several datasets, containing a total of well over 1,000 patients, *AML-BET* can perform differential gene expression and survival analyses and return detailed visualizations of the analyzed data. *AML-BET* will not only maintain a database of genes and their expression data across several datasets but will also process and normalize its data through *TMM*, one of the most effective methods available, distinguishing it from other tools. These analyses and visualizations will allow for the discovery and evaluation of candidate biomarkers, which may prompt future research into potential treatments specific to the biomarker.

1. Introduction

Significance

Cancer is a category of disease in humans and other organisms, derived from an individual's genetics, that is caused by mutated or abnormally expressed genes. Gene expression can directly correlate with an increased or decreased risk to develop or succumb to cancer, as an increased level of the RNA or protein associated with that gene may increase or decrease the risk factors of an individual. We use gene expression profiling to measure the expression levels of genes to investigate cancer, granting us access to data that can reveal prognostic and diagnostic biomarkers.

Prognostic biomarkers are molecules that are indicative of disease outcome; diagnostic biomarkers are molecules that can be used to diagnose or identify the existence, stage, grade, or clinical subtype of cancer (Kumar et al., 2006). It is these biomarkers specifically that we are interested in, because, due to the nature of cancer being a genetic disease, cancer is very personalized and specific to the patient. Through the identification and analysis of candidate biomarkers, clinical tests could be created allowing for specialized and more effective treatments.

Acute Myeloid Leukemia (AML) is a form of leukemia that starts in the bone marrow, affecting blood-forming cells called myeloid cells, hence the name. AML is the most common form of leukemia in adults, and is the most aggressive cancer, with a wide possibility of prognoses depending on the specific kinds of molecules that make up the cancer

(Chennamadhavuni et al., 2024). Across the different subtypes of AML, there is a large amount of variation. For example, the 5-year net survival rates between different AML subtypes range between 9% and 68% in men (Mounier et al., 2021). This level of variation requires treatment plans to be specialized, as the many subtypes of AML are highly heterogenous in nature (Kumar et al., 2023).

It is for these reasons that we are so interested in gene expression data. Using gene expression datasets for AML patients, we can find associations between the expression of certain genes and the likelihood of succumbing to AML or any of its subtypes. Through these associations, genes with higher or lower expression can be identified as biomarkers, which would then allow clinicians to provide their patients with more in-depth treatment plans based on the patient's exhibition of these biomarkers.

Currently, while there are many gene expression and AML-centric databases, there are no online databases or web services that allow for the automated and comprehensive evaluation of candidate diagnostic and prognostic gene expression biomarkers in patients across multiple datasets. The objective of this thesis is to remedy that issue and create *AML-BET* (Acute Myeloid Leukemia Biomarker Evaluation Tool), an AML-specific database and web application that allows users across the Internet to easily access and analyze genes without requiring an extensive bioinformatics education or in-house tools and analyses.

Background

Gene expression and cancer

DNA is the blueprint from which every cell in the human body takes its instructions from. Genes are comprised of DNA, which are either active or inactive. In general, if a gene is active,

its genetic information will be transcribed into mRNA. Protein synthesis then involves the translation of this mRNA within a cell, allowing the DNA to guide the functioning of the human body by contributing behavior-inducing protein at a cellular level (Urry et al., 2021).

The same DNA exists within every cell inside the human body. Each of those cells may be subject to a mutation of the genes within their copy of DNA. If this occurs, the consequent RNA produced can be altered, representing a mutated gene instead of the intended gene. This process can directly affect how the cell functions, as the cell may either cease production of a protein that assists in homeostasis or may alter the production of a protein that actively harms homeostasis. This could be as simple as the overproduction of a protein central to cell division, causing out-of-control cell division and growth, which is a hallmark of cancer (Hanahan and Weinberg, 2011).

While the mutations a cell can experience may be simple, there are mutations that directly alter the way a patient responds to treatment. For example, a mutation could cause the production of a protein that increases a cell's resistance to cytotoxic (*toxic to cells*) treatments, such as chemotherapy. In this case, chemotherapy, one of the main treatments used to combat cancer, is likely to have a significantly lessened effect. This is one reason that studying and analyzing genomic data has an important role in cancer treatment: if we can identify the specific genetic factors that are contributing to the proliferation and survival of a patient's instance of cancer, we will be able to treat them much more effectively (Dancik et al., 2023).

Additionally, a precedent for this kind of genomic data analysis has already been established. MammaPrint is a genomic test that analyzes the expression of 70 genes in early-stage breast cancer, which predicts outcome. With the results of the MammaPrint testing, the physicians of the patients within the testing group found an increase in the confidence of 72.2%

of their treatment plans, with 88.5% of treatment plans aligning with the results of MammaPrint (Soliman et al., 2020). It is this additional information from biomarker evaluation that we wish to pass on to physicians, allowing patients to receive much more specialized and effective care.

We can identify the specific genetic factors of a patient's cancer through analyzing their gene expression – the amount of RNA being produced from any given gene. The greater the amount of RNA being produced from any given gene, the greater the number of proteins associated with that RNA are produced. Proteins directly affect the behavior of a cell. Therefore, the higher the gene expression, the higher the impact on the cell's behavior will be. Expression of a mutated gene or an abnormally high or low expression of a wild-type gene may then correlate to the presence of cancer cells, or to specific characteristics of a cancer. This allows us to identify candidate genes, like the example gene providing cytotoxic resistance, as biomarkers indicating a specialized treatment for the patient. What makes a gene notable, or a possible candidate biomarker, is not just raw expression either; it is the difference in the expression of that gene between normal and tumor samples, or samples with distinct clinical characteristics. Many genes simply have a high expression by default- what would be significant, then, is a sizable difference in the level of expression of that gene in a cancer cell with the same active gene.

Cancer, as a category of disease, is dependent on the individual genetic makeup of an individual, and any acquired genetic changes, such as mutations from external carcinogenic substances. The genes residing within a person's genome, as well as which variants, how they mutate, and how much they are expressed, dictate both the characteristics and severity of that specific instance of cancer. Even if two people develop the same type of cancer, such as AML, it is often more useful to think of those two people as having two different diseases. Cancer is dependent on their genetics – something that can be unique to them.

Acute Myeloid Leukemia

Leukemias are a form of cancer that starts in cells that develop into different types of blood cells. They can begin with either myeloid cells or lymphoid cells, both of which are found within the bone marrow of a patient. Leukemia can also either be acute or chronic, which is fast-growing and slow-growing, respectively. The designation of acute vs. chronic leukemia is determined by the percentage of immature cancerous blood cells, known as blast cells, as acute leukemia replicates so quickly the cancer cells are often left immature and underdeveloped. The diagnosis of acute Leukemia requires a blast percentage of over 20 ("Tests for Acute Myeloid Leukemia (AML)," n.d.). Acute Myeloid Leukemia (AML) is a fast-growing, myeloid-originating leukemia that starts in the bone marrow of a patient and quickly spreads into the blood, as well as other organs related to blood filtration ("What Is Acute Myeloid Leukemia (AML)?," n.d.).

AML is the most common form of aggressive leukemia in adults, comprising 34% of all leukemia diagnoses ("Overview of Leukemia - Hematology and Oncology," n.d.). The annual incidence of new cases in both men and women is ~4.3 per 100,000 population, with over 20,000 new cases per year in the US alone. The median age of diagnosis is 68, with a higher incidence in males compared to females, with a ratio of 5:3 (Vakiti et al., 2024). The five-year survival rate of AML has a very large range (as seen above), but most often resides below just 30% across all subtypes and ages (Mounier et al., 2021). Childhood leukemia, while predominately Acute Lymphocytic Leukemia (ALL), accounts for almost 33% of all cancers in children ("Leukemia in Children | Childhood Leukemia," n.d.). The rest of the cases of Leukemia among children are AML.

The World Health Organization (WHO) has designated multiple different subtypes of AML, which are organized by the gene mutations associated with them and the percentage of blast cells within samples (Döhner et al., 2022). These distinctions, based on the genetic aberrations of an individual, are required due to the large impact they have on the expression of the disease and the outcome of the patient. The European LeukemiaNet (ELN) is a European transnational group dedicated to curing Leukemia, and they have designated risk classifications for each of the subtypes created by the WHO. These risk classifications fall into Favorable, Intermediate, and Adverse (Döhner et al., 2022). These risk classifications are sourced from the genetic subtypes as designated by WHO; it is the genetic makeup of the leukemia that directly determines the risk of the disease. *AML-BET* aims to assist in the identification of candidate biomarkers to further understand the genetic makeup of leukemia subtypes. It is through this deeper understanding that more effective treatments for these highly variable subtypes can be developed.

The importance of subtyping by genetic aberrations cannot be understated. AML subtypes are derived from a large variety of different genetic mutations, the effects of which can change the phenotype and outcome of the disease drastically (Döhner et al., 2022). The differing phenotype and outcome affect more than just risk; many of them require entirely different treatment plans and medications. Similar to the effect MammaPrint has on treatment efficacy and physician treatment confidence (as described above), biomarker evaluation in AML samples would allow physicians to more accurately and effectively treat their patients.

Gene Expression Analysis

To evaluate candidate biomarkers through gene expression analysis, we can first obtain the gene expression data from an individual's AML cell samples as compared to their "normal" cell samples. One method of performing such analysis, DNA microarrays, involves a chip that must be manufactured and programmed to detect specific genes. These chips contain small "wells", each of which contains a DNA probe that corresponds to a specific gene or gene mutation. Normal and tumor samples of isolated mRNA are taken and are converted to complementary DNA (cDNA) ("DNA Microarray Technology Fact Sheet," n.d.). cDNA contains the gene of the mRNA as well as a fluorescent dye to distinguish the two samples. Both normal and tumor cDNA from the same individual are mixed and added to the microarray. The wells then represent either sample by displaying a color, corresponding to the dyes used earlier ("DNA Microarray Technology Fact Sheet," n.d.). If a well is representative of the normal cDNA's color, the normal sample has a higher expression of that gene. If it is representative of the tumor's cDNA color, the tumor sample has a higher expression of that gene. If a well represents a mixture of the two colors, the samples likely have a similar expression of that gene. This same analysis can be performed on single samples (e.g., only tumor samples) in which case the well is represented as an intensity of a single color corresponding to the expression of the gene. While microarrays can be useful for determining the presence of specific genes or gene mutations, as well as a relative idea of their compared expression, they do not detect unknown genes or mutations and are not able to be adjusted or modified post-manufacture.

RNA Sequencing (RNA-Seq) is a more flexible approach to determining gene expression. RNA-Seq takes isolated mRNA and breaks it down into fragments, which are sequenced to yield "reads", which correspond to portions of the mRNA samples being analyzed. The individual genes being represented by the reads are determined through the use of a reference genome. The

reference genome has introns (non-coding portions) removed, and the remaining sections are separated into genes. The read alignment process then occurs, where the reads are mapped onto the reference genome, showing the number of reads that correspond to each gene in the genome (Kukurba and Montgomery, 2015). After the reads are aligned to the genome, we determine the apparent expression of each gene through quantification, which is the number of reads present for a gene. In general, the greater number of reads, the higher the expression of that gene.

While RNA-Seq provides a much more efficient method of determining gene expression, it faces the issue that read counts are affected by differences in sample sizes between individual samples. Sequencing depth is another way to refer to the total number of reads found in a sample. While read counts do measure expression, if a tumor sample is larger than a normal sample, and both have the same relative expression of a certain gene, the sequencing depth of the tumor sample will be significantly larger than the sequencing depth of the normal sample, showing a greater gene expression when there is none (Piper, 2017). Some genes may also have more reads than others due to their longer length, as a gene with a longer length will have more base pairs for potential reads to map to (Piper, 2017). RNA composition is also something to take into consideration, as if we are comparing the same gene between two samples, but one sample has a much higher expression of an unrelated gene, the read counts of the gene of interest will be influenced by the unrelated gene. These issues are remedied through gene expression analysis methods to normalize the data, of which we will cover two: FPKM/RPKM and TMM.

F/RPKM (Frames/Reads Per Kilobase Per Million) is a commonly used normalization method that attempts to address the sequencing depth issue by normalizing the read count based on gene length and the total number of mapped reads ("FPKM - GDC Docs," n.d.). This is done by multiplying the number of reads mapped to a gene (RM_g) by a scalar of 1 million and dividing

it by the total number of mapped reads across the entire read-alignment reference genome (RM_t) multiplied by length of the gene in question (L). This creates the formula $FPKM = \frac{(RM_g \times 10^9)}{(RM_t \times L)}$.

This method, while concise and efficient, has a glaring issue, because the FPKM value of a gene can still be impacted by unrelated genes. In any given method, an acceptable false-positive rate is set by the analyst, and is typically 0.05, or 5%. Analysts use a 5% false-positive rate as a standard with which to perform and build their analysis around. Power, in an analysis method, refers to the effectiveness or sensitivity of any given method. The higher the power, the more likely you will return true positive results for your query. With FPKM, under any level of power, the range for false positive rates for detecting differential genes is 10% to nearly 20%, despite a specified false positive rate of 5% (Dillies et al., 2013). This range is not acceptable for a biomarker evaluation tool, and performing analyses with an unexpected false-positive rate this high leads to high expenses and many difficulties. *AML-BET* will be implementing a different form of normalization, TMM, which is described next.

TMM (Trimmed Mean of M Values) is a much more accurate normalization method. It begins by determining the fold-change (proportion or ratio) of each gene across samples being analyzed, with respect to a reference sample. These values will become our "M" values. We want to perform our normalization with the assumption that most genes are not differentially expressed, so we want our fold-change to be as close to 1 as possible. This will make the presence of differentially expressed genes much more apparent. We achieve this by "trimming" or removing the upper and lower 30% of the M values, after trimming the top 5% of the genes with highest expression. This leaves us with only the M values of the most equally differentiated genes, which we then take the average of. This final averaged value becomes our normalization

factor, or effective sequencing depth, which we apply to the respective samples to receive our TMM value of gene expression (Robinson and Oshlack, 2010). TMM, as opposed to FPKM, provides a near-constant false-positive rate of 5%, the acceptable rate, under any level of power (Dillies et al., 2013). This form of normalization is one of the most effective and accurate methods and will be the form utilized for RNA-Seq data in *AML-BET*.

Relevant Software

For the execution and visualization of gene expression analyses, *R* will be used. *R* is a data analysis/statistics-focused programming language. It is interpreted, and is most often used to compile, compute, and display large amounts of data in an intuitive and efficient manner. *R* has bundles of code and functions created by the community known as *packages* that can be directly inserted into the library of an *R* workspace. These packages can improve on standard *R* processes, but can also perform actions outside of the original scope of *R*. One such package, *Shiny*, can be used to generate HTML and to develop webpages. This feature will be used to develop *AML-BET*, as well as supply the webpage with the visualizations and analyses carried out and produced by *R*. *Shiny* also includes all the formatting and features that come with HTML, such as CSS styling and light scripting capabilities.

A downside of using *R* and *Shiny* to host a web application is the fact that *R* is a single-threaded language. It is impossible for *R* to carry out more than one thread of control or computation at once, so any individual connecting to the application would have to wait in a queue to be served by the program. *ShinyProxy* is an application that uses *Docker* images and launches *Shiny* apps from them, as opposed to a single instance of the app running alone in the

Docker image. This allows multiple instances of *R* to be launched, allowing for several threads to service several users at once, instead of sharing an individual thread. *Docker* is a lightweight form of a virtual machine running an operating system with only the required content to maintain a web application, such as *R*, *Shiny*, and *ShinyProxy*.

For the database back-end, instead of using data frames read directly into *R*, *MongoDB* will be used to store and query the required data for *R* to generate visualizations and perform analyses. *MongoDB* is a non-relational, document-based database that scales well and requires much less initialization time than a traditional or SQL-based database. Documents are stored as JSON files and can contain a multitude of data types (including other documents) that are easy to insert and manage. *MongoDB* was chosen as the database of choice for this application due to its simplicity and efficiency, as well as *AML-BET*'s lack of a need for relationships, writing, or constant upkeep.

Literature Review of Related Bioinformatics Tools and Databases

Of the list of already-existing bioinformatics tools, *BC-BET* (https://gdancik.github.io/BC-BET/) is the closest in design and analysis to the proposed *AML-BET*. It will function as a loose template for the development of *AML-BET*, alongside advice from the developer of *BC-BET*, Dr. Garrett Dancik. *BC-BET* was created to evaluate candidate diagnostic and prognostic biomarkers regarding genes related to bladder cancer (Dancik, 2015). *BC-BET* allows for the evaluation of individual genes or multiple genes and designates samples by their grade and the muscle-invasiveness. Analyses within *BC-BET* include tumor, grade, stage, and survival (which can be sorted by muscle-invasiveness) for the gene or genes selected. *BC-BET*, being a bladder cancer

evaluation tool, lacks the AML-centric features and datasets required for the evaluation of biomarkers in AML. *AML-BET* will be able to perform analyses across AML risk groups.

UCSC Xena (https://xena.ucsc.edu/) is an online genomic data analysis tool with a focus on ease-of-use and clarity. The site operates by prompting the user to pick a singular dataset from which to pull genomic data, and then asks for 2 or more variables to visualize from the dataset. These variables can be either genomic or phenotypic, allowing a user to specify certain genes, their expression, and other clinical data like "days to death" or sample site. One of the main drawbacks of using UCSC Xena is its limitation of only one dataset per analysis. This limits the ability of a user to draw conclusions, as anything they perform with one dataset must be performed separately in another. Further muddling this aspect of UCSC Xena is the potential risk of these datasets not being uniform in their format or content. Without the processing of these datasets in ways specific to their disease that allow their analyses to be consolidated and used more than one at a time, UCSC Xena is limited in its ability to efficiently carry out genomic data analysis for biomarker evaluation. Finally, the gene expression data is only represented in FPKM (or similar methods), a normalization method with a nearly 20% false-positive rate, which calls into question any results based on comparison between samples (Dillies et al., 2013).

cBioPortal (https://www.cbioportal.org) is an online genomic evaluation tool with a focus on gene mutations. While gene expression is a factor in some of the analyses available, such as Kaplan-Meier survival curves or co-expression between a gene and a mutation, the greater focus of the program is on individual gene mutations. There exists no way on *cBioPortal* to draw a correlation between the expression of a gene between two subtypes of AML, as the database does not designate the samples by subtype or risk. *cBioPortal* also uses FPKM (or related methods) to calculate and display gene expression data, which has a nearly 20% false-positive rate (see above

for details). While the graphing capabilities of cBioPortal are diverse, they do not cover in-depth comparisons of gene expression between the different forms of AML mentioned above. The main goal of *AML-BET* is to provide these in-depth analyses to assist in biomarker evaluation, and the significance of these analyses cannot be understated. The user interface of *cBioPortal* is readable and intuitive, and its UI and UX should be emulated.

KM Plotter (https://kmplot.com/) is a web tool specifically designed to produce detailed Kaplan-Meier plots visualizing survival rates across several different cancers. Within the AML *KM Plotter*, there is a very high level of specialization, allowing you to limit data used by AML subtype, the gender of the subject, where the sample was taken from, as well as the treatment the subject received. Datasets can be considered individually or all at once when creating plots, and the tool tells you your total sample size under any given configuration. AML datasets used in *KM Plotter* are limited to microarray datasets, and do not include any RNA-SEQ datasets. Genes are selected by the user, with 100 being available to graph at one time. Samples can be divided between "high" and "low" expression to define the groups of any given Kaplan-Meier plot, or the "best cutoff" can be selected. *KM Plotter* is unparalleled in its generation of Kaplan-Meier survival graphs, and its success will be emulated by the survival analyses by gene expression in *AML-BET*.

The *Leukemia Gene Database* (*LeGenD*; http://www.bioinformatics.org/legend) is a database site run by Bioinformatics.org. The database includes a comprehensive list of genes associated with the many forms of leukemia. The entries are labeled by each gene's chromosome number, gene name, and gene loci. Within the entry, each gene has a library of genotypic and phenotypic data sourced from studies, with each claim or piece of information being followed by a citation. The intention of the site is to be used as the ultimate reference guide for genes

associated with leukemia. If you have questions regarding a gene, simply look it up in the database and retrieve a comprehensive list of research publications regarding it. This site, while useful, contains no gene expression data, analyses, or visualizations of any kind. It does, however, deliver concise and comprehensive data in an easy-to-read and easy-to-access manner, and its success in this should be emulated.

3. Methodology

The datasets that will be included in *AML-BET* will come directly from a list of datasets that were used to determine the effectiveness of the ALDH family of genes as candidate biomarkers in a study carried out by Dr. Garret Dancik (Dancik et al., 2023). These seven total datasets, with 1,170 patients, include both RNA-Seq and microarray data from the *Gene Expression Omnibus* (*GEO*), the *Genomic Data Commons*, and the previously mentioned *cBioPortal*. Additional datasets will be considered as well during the development of *AML-BET* and will be added if they are found to be sufficient.

RNA-Seq datasets will be processed using TMM and will also include several genotypic and phenotypic variables with which to graph against the expression data. Visualizations include boxplots comparing expression across different groups of samples (such as patient risk groups), Kaplan-Meier survival plots, and forest plots.

MongoDB will be used as the database back-end for *AML-BET*, with datasets formatted and processed in *R* before being inserted into the database. The datasets will be formatted to maintain a level of uniformity between the datasets to make analyses easier for the user to carry out. *Shiny* will be used to generate HTML complete with embedded data visualizations and other

details, as well as CSS styling and scripting. A *Docker* image will be used to host *AML-BET*, and *ShinyProxy* will be installed to allow for our web application to mimic multi-threading and increase serviceability. A timeline for the development of *AML-BET* and the completion of the final thesis is listed below in Table 1.

Table 1. Timeline of Proposed Research

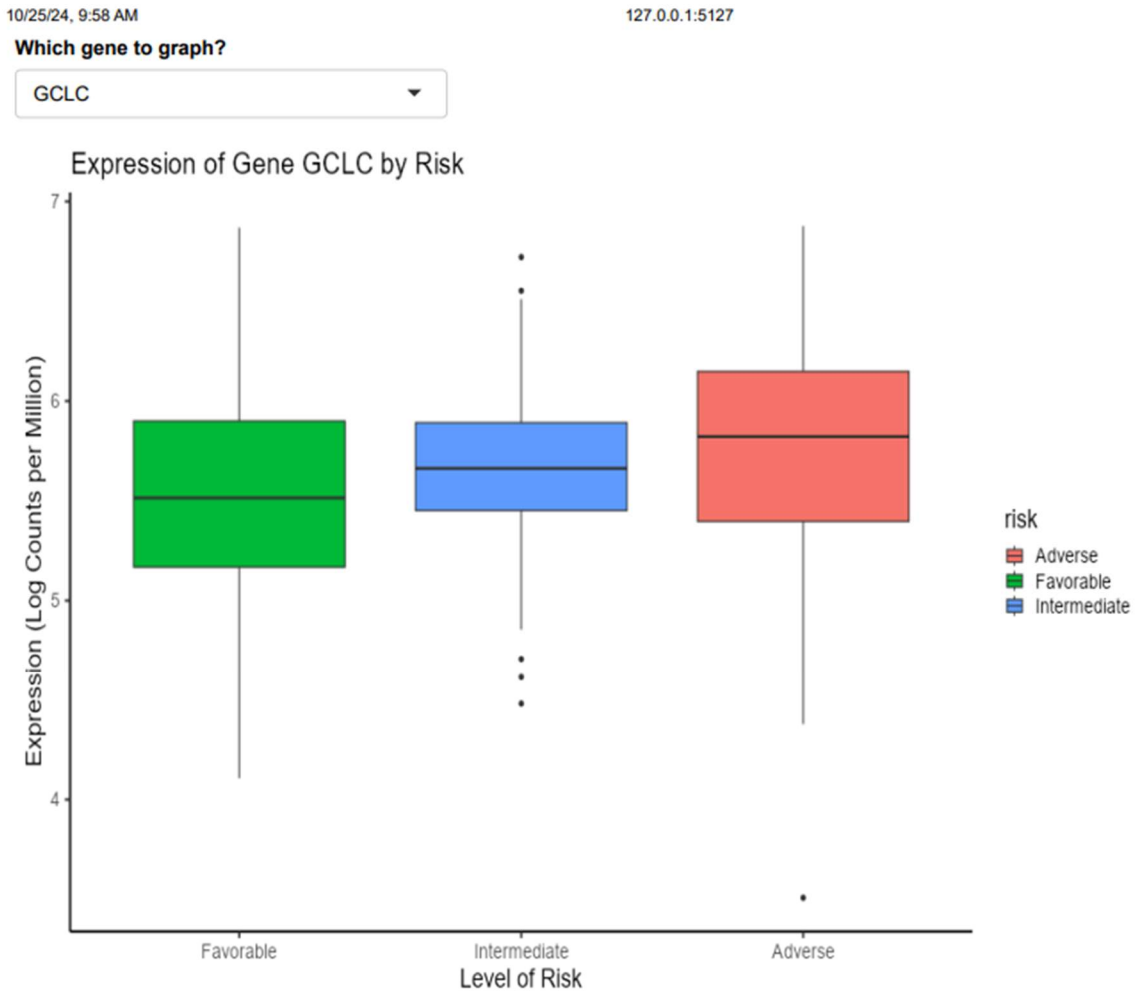
Deadline:	**Step:**
March 17th, 2025	Finish processing and transferring datasets into MongoDB.
May 31st, 2025	Finish Homepage of *AML-BET*, complete with all essential functions.
September 31st, 2025	Finish development of *AML-BET*, complete with all optimizations and quality-of-life features.
October, 2025	Present at COPLAC Northeast Regional Conference.
November 15th, 2025	Turn in thesis to second reader.
December 4th, 2025	Turn in thesis to Honors Director.
April, 2026	Present at NCUR Conference.
April, 2026	Present at CCSCNE Conference.

4. Preliminary Work

Several pieces of preliminary work, whether they be classes, independent studies, or assignments, are in progress or have been completed in order to develop a strong background in bioinformatics and to prepare myself to create *AML-BET* and complete this thesis. In CSC 314, "Introduction to Bioinformatics", the basics of genetic biology and genomic data analysis were covered. The final assignment entailed developing a local dynamic programming matrix in a programming and was written in Java. This class was followed by an independent study in CSC-315, "Genomic Data Analysis", which is in progress, and which covers more in-depth concepts surrounding bioinformatics. Properly using *R*, processing and visualizing data, understanding RNA-Seq, and applying TMM to gene expression data were all covered in thorough detail.

To prepare for the development of *AML-BET* regarding the web application's reliance on *R*'s *Shiny* package, a small *Shiny* application was developed to display gene expression data by the risk groups assigned to each patient. The app reads in the BEAT_AML dataset, one of the datasets that will be used in *AML-BET* and stores its information in a list. A function then takes in two arguments: the dataset, and the string that corresponds to the gene name being graphed. When a user selects one of 10 sample genes, the app runs the function to extract all expression data related to the gene and adds in all patient risk data accordingly. This data is then output in the form of a side-by-side boxplot and displayed to the user, which can be seen below in **Figure 1**. This app may also act as a "prototype" of sorts; this application is representative of the functionality and one form of analysis that will be present in *AML-BET*, although limited to one dataset and 10 genes.

Figure 1. Gene Expression by Risk Shiny Application



5. Expected Outcomes

The result of this thesis will be the finished development of *AML-BET*. *AML-BET* will be capable of rendering custom visualizations of gene expression data analysis through several forms, such as Kaplan-Meier survival plots, boxplots, and forest plots. The web application will allow the user to pull data from several datasets at once, as well as generate visualizations that

filter the data to draw distinctions between different clinical traits of the patients in the datasets. A user will have the ability to form conclusions regarding differential gene expression based on risk, dataset, sex, age, and other clinical details. In addition, *AML-BET* will be submitted to the Honors Council in the form of an honors thesis and will be presented publicly at conferences such as at the CREATE conference, as well as others related to the field of bioinformatics.

6. Conclusion

AML-BET will take the form of an always-available web application, running through *ShinyProxy* installed into a Docker image. When a user connects, they will be given their own R instance with which to perform visualizations, and many users will be able to simultaneously use the web application at any given time. *AML-BET* will contain a graphical user interface that is intuitive and easy to operate, as well as capable of accurately and efficiently supplying data and visualizations to the user. Any newly generated datasets, such as those created by a user performing differential expression over a variable, will be available for download as an image or .csv file for academic use.

The ultimate end goal of *AML-BET* is to assist in clinical research regarding the genomic data of Acute Myeloid Leukemia. Users will have the opportunity to find sufficient data to determine a gene to be highly associated with some aspect of AML. This data can then be used to generate further research into the gene and its traits, potentially even leading to new drugs or treatment plans being made for individuals with a specific expression of that gene. Without a tool like *AML-BET*, one dedicated to the discovery of these associations, the clinical researchers studying AML would have to either perform interdisciplinary, high-level genomic data analysis,

or consult with a bioinformatics collaborator to obtain or analyze their data. *AML-BET* allows those without a background in computer science or bioinformatics to perform analyses on their own, increasing the availability of this data, as well as the likelihood that more advances are made into better understanding and treating AML.

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